Algorithme V_0

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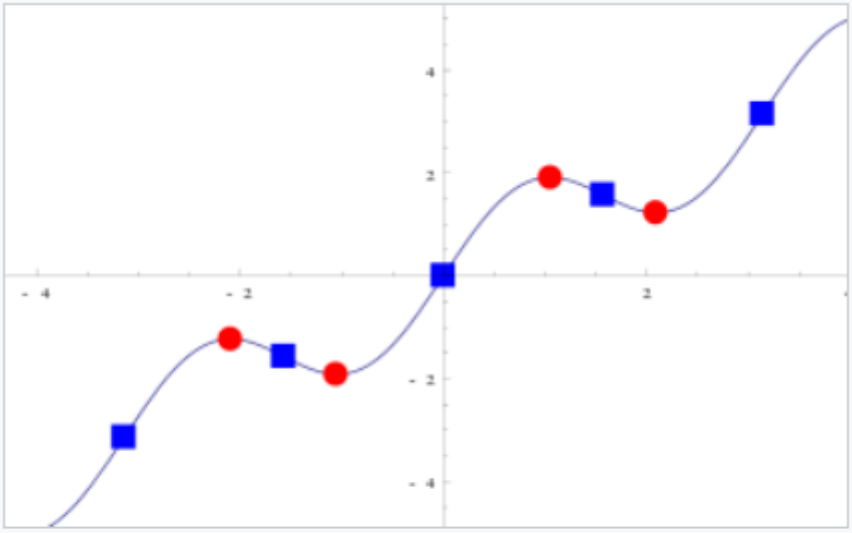
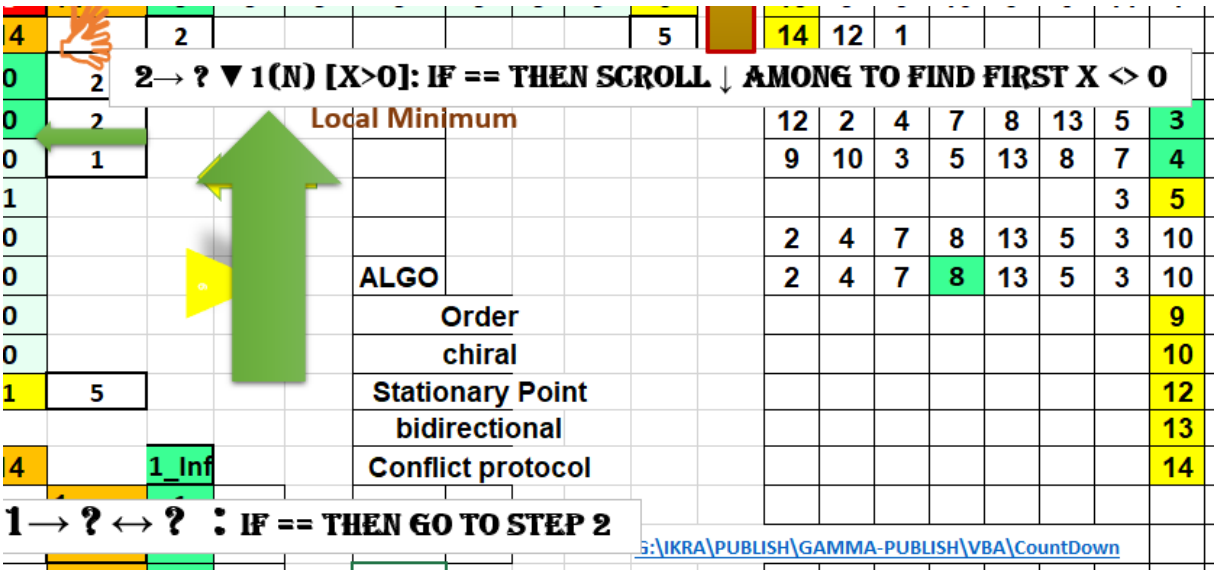


Les points stationnaires de la fonction sont marquées par des ronds rouges. Dans ce cas, ce sont des extrema locaux. Les carrés bleus désignent les points d'inflexion.



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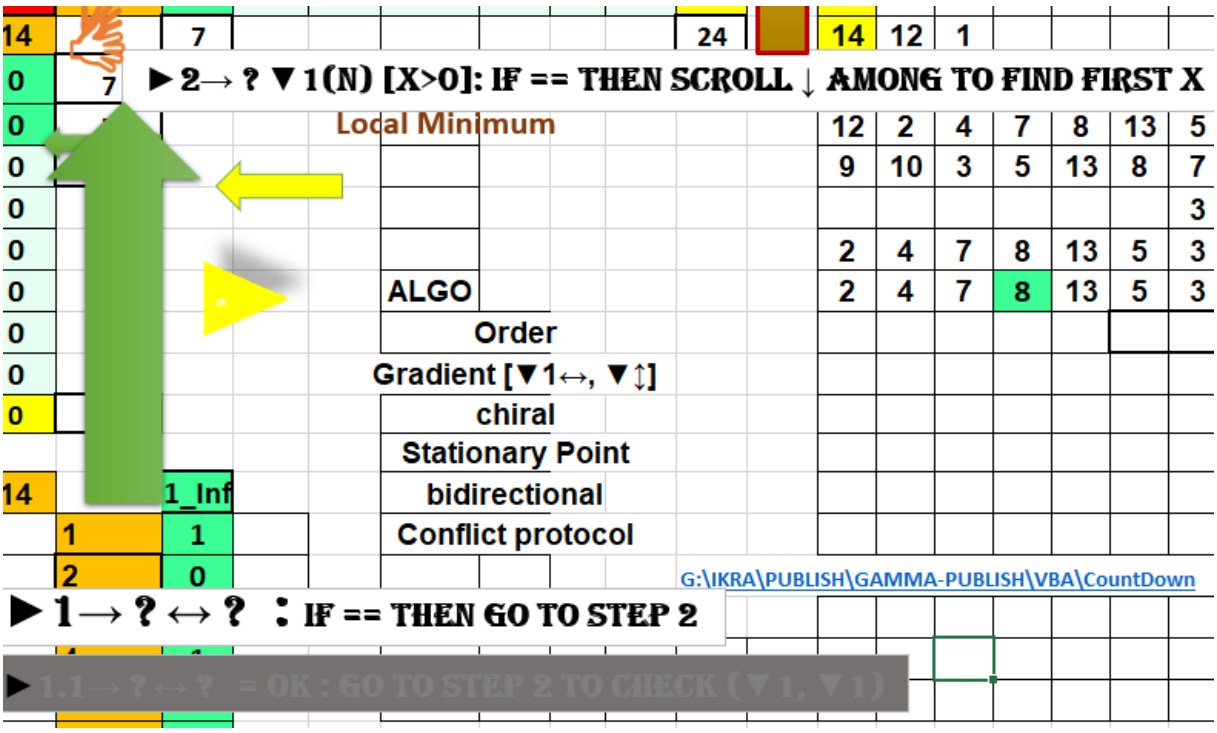

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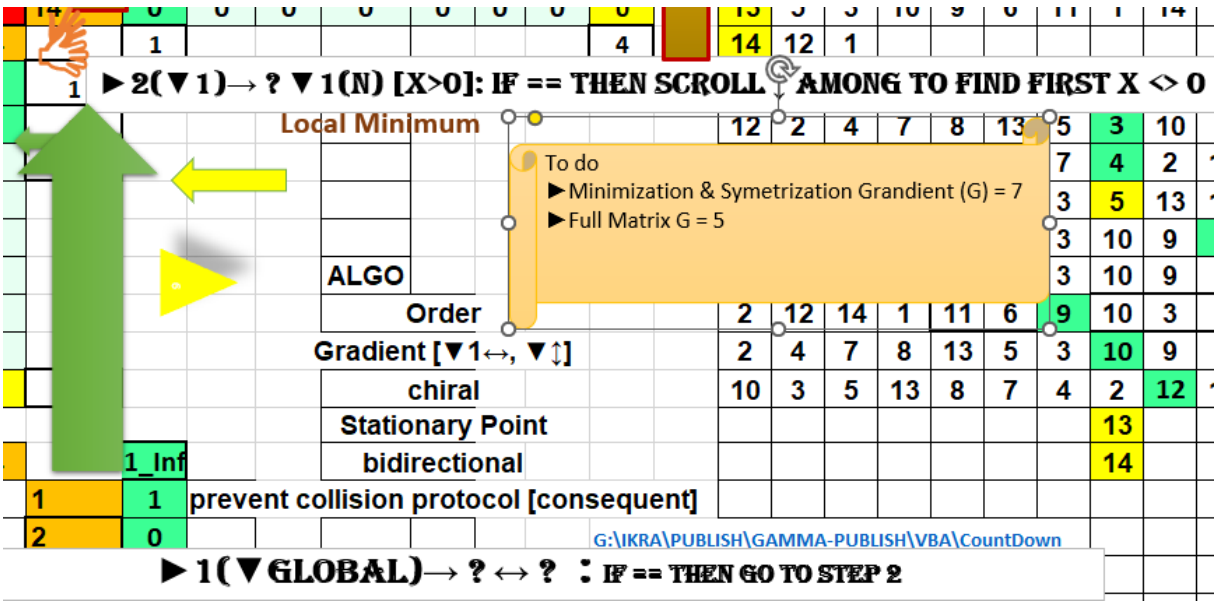

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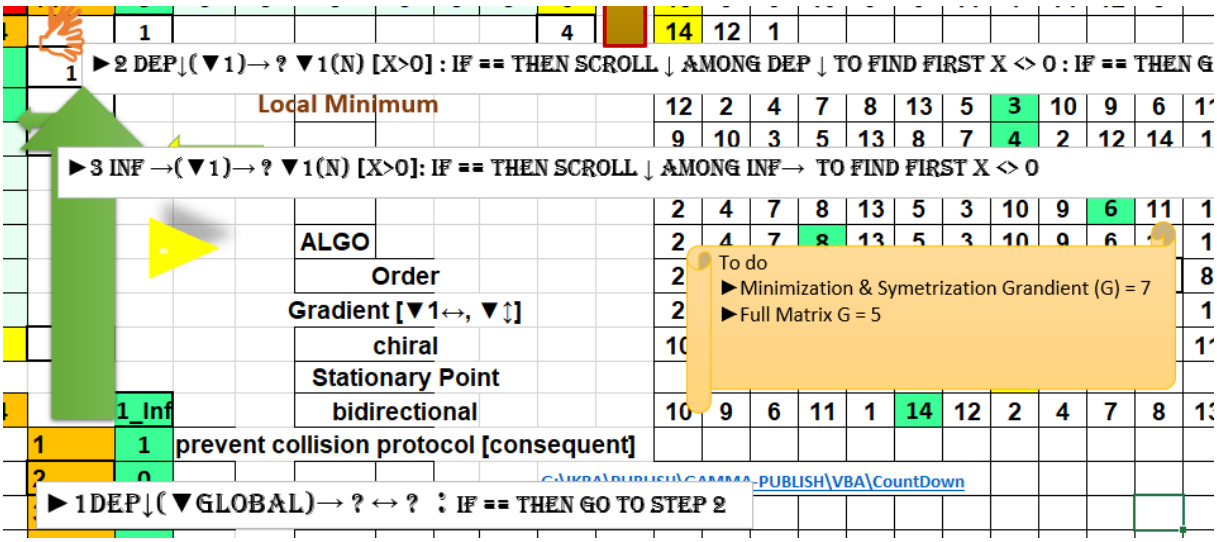

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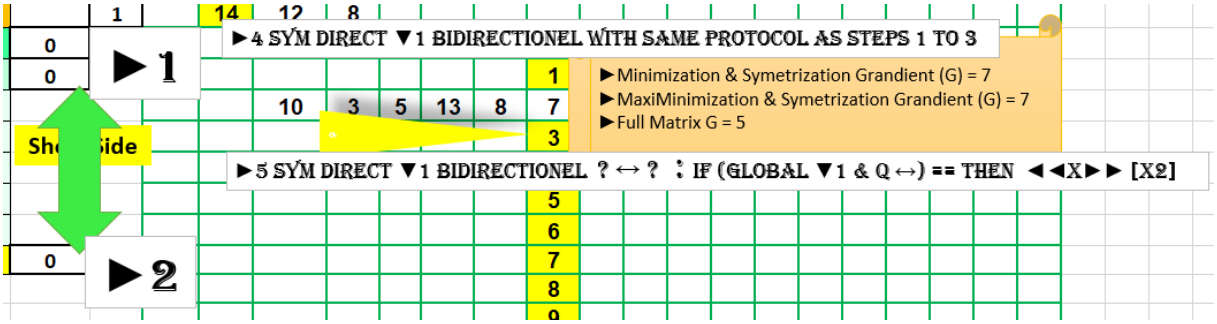
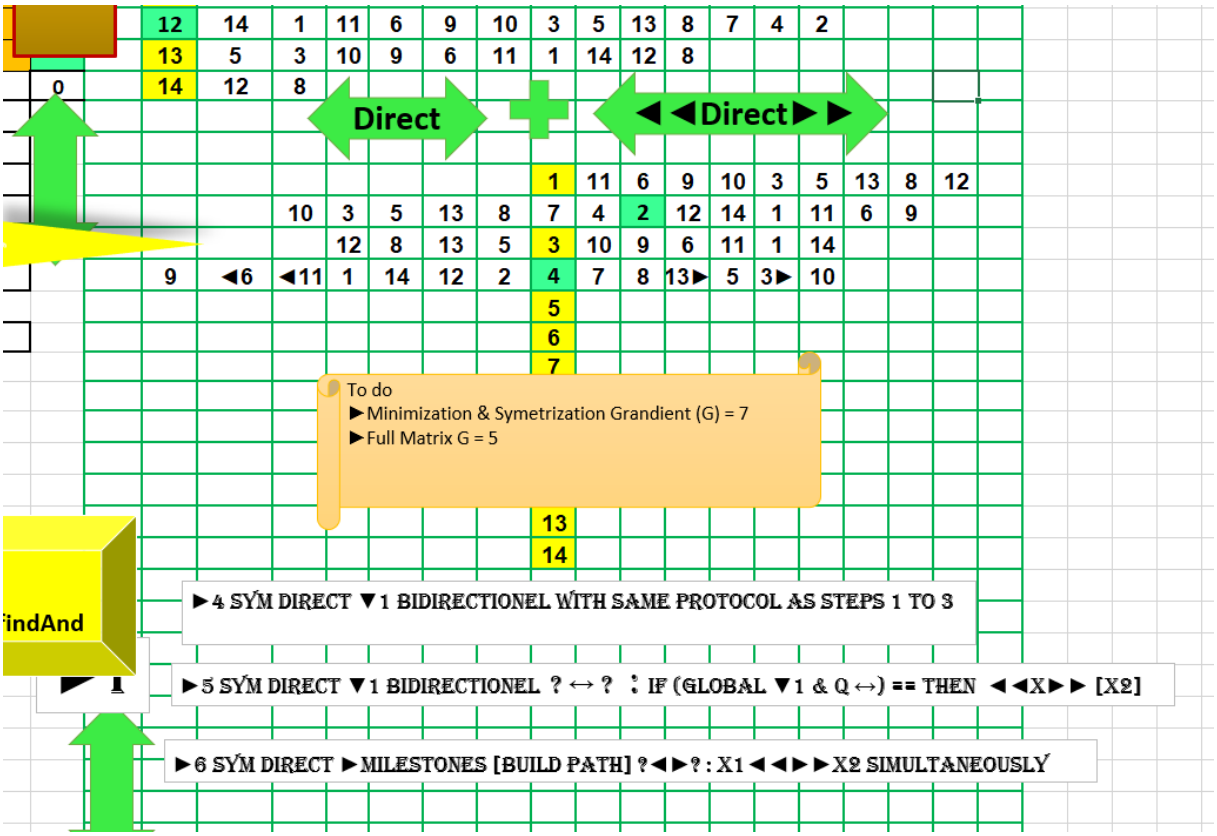
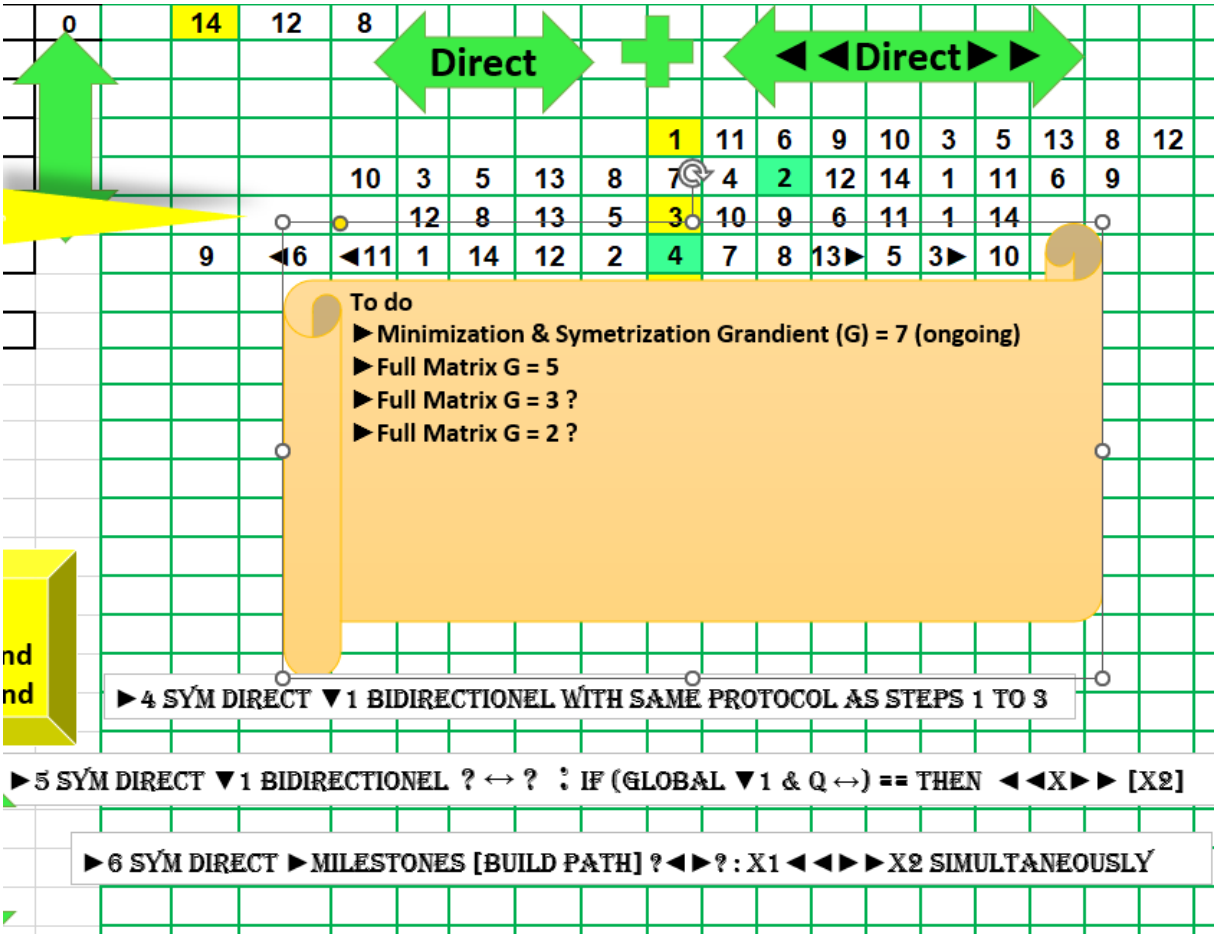


02/05/2023 17:35:21


Direct
◄◄Direct►►
To do
►Minimization & Symetrization Grandient (G) = 7
►Full Matrix G = 5
►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS STEPS 1 TO 3
►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN ◄◄X►► [X2]
►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2 SIMULTANEOUSLY

02/05/2023 17:49:03



02/05/2023 19:06:41

find
And

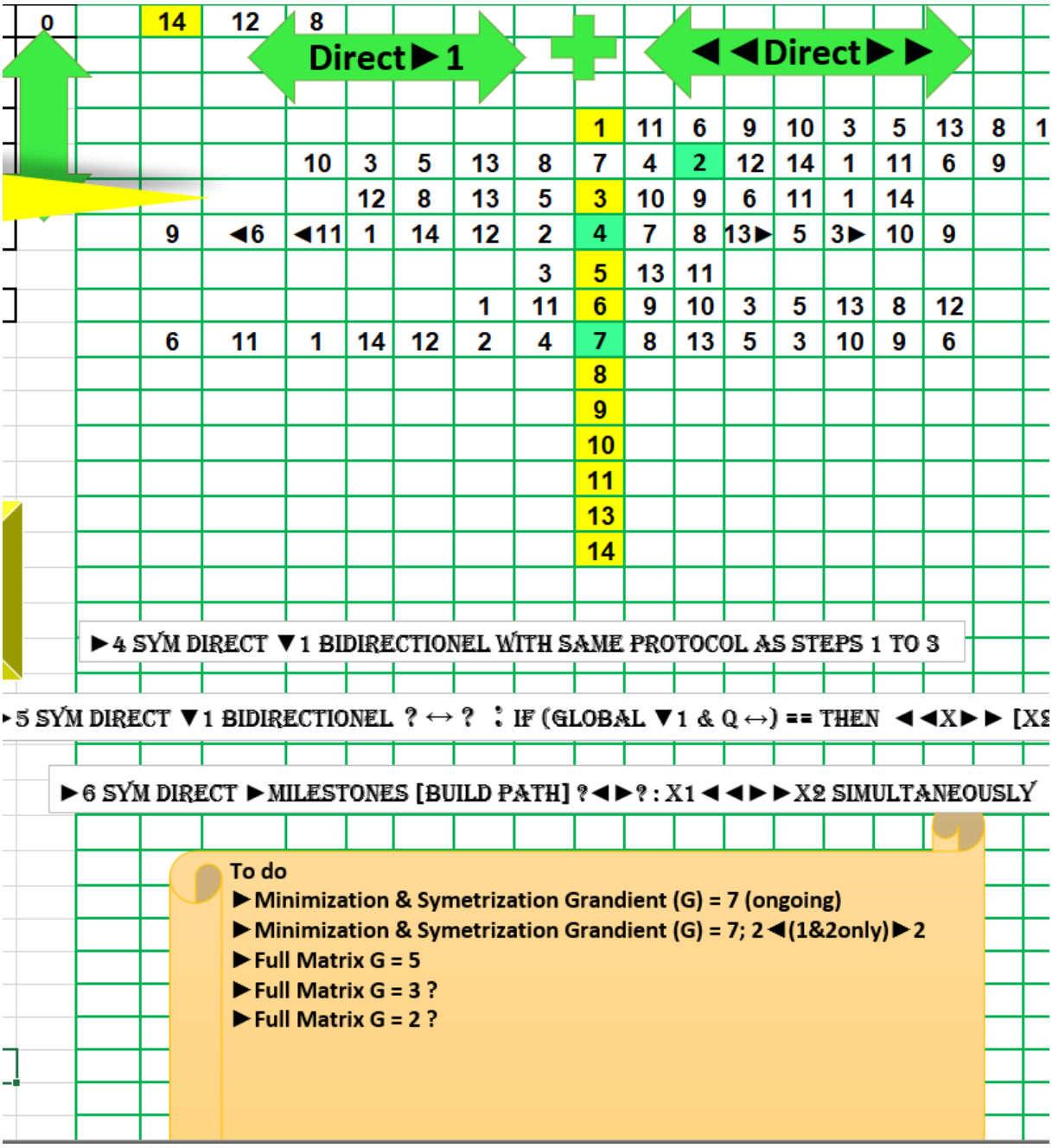
►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS STEPS 1 TO 3

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN ◄◄X►► [X2]

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1 ◄◄►►X2 SIMULTANEOUSLY

To do
►Minimization & Symetrization Grandient (G) = 7 (ongoing)
►Minimization & Symetrization Grandient (G) = 7; 2◄►2
►Full Matrix G = 5
►Full Matrix G = 3 ?
►Full Matrix G = 2 ?

02/05/2023 19:30:12



02/05/2023 22:05:52

	13	5	3	10	9	6	11	1	14	12	8						
0	14	12	8														
								1	11	6	9	10	3	5	13	8	12
			10	3	5	13	8	7	4	2	12	14	1	11	6	9	
				12	8	13	5	3	10	9	6	11	1	14			
	9	◄6	◄11	1	14	12	2	4	7	8	13►	5	3►	10	9		
							3	5	13	11							
						1	11	6	9	10	3	5	13	8	12		
	6	11	1	14	12	2	4	7	8	13	5	3	10	9	6		

Direct►1

◄◄Direct►►

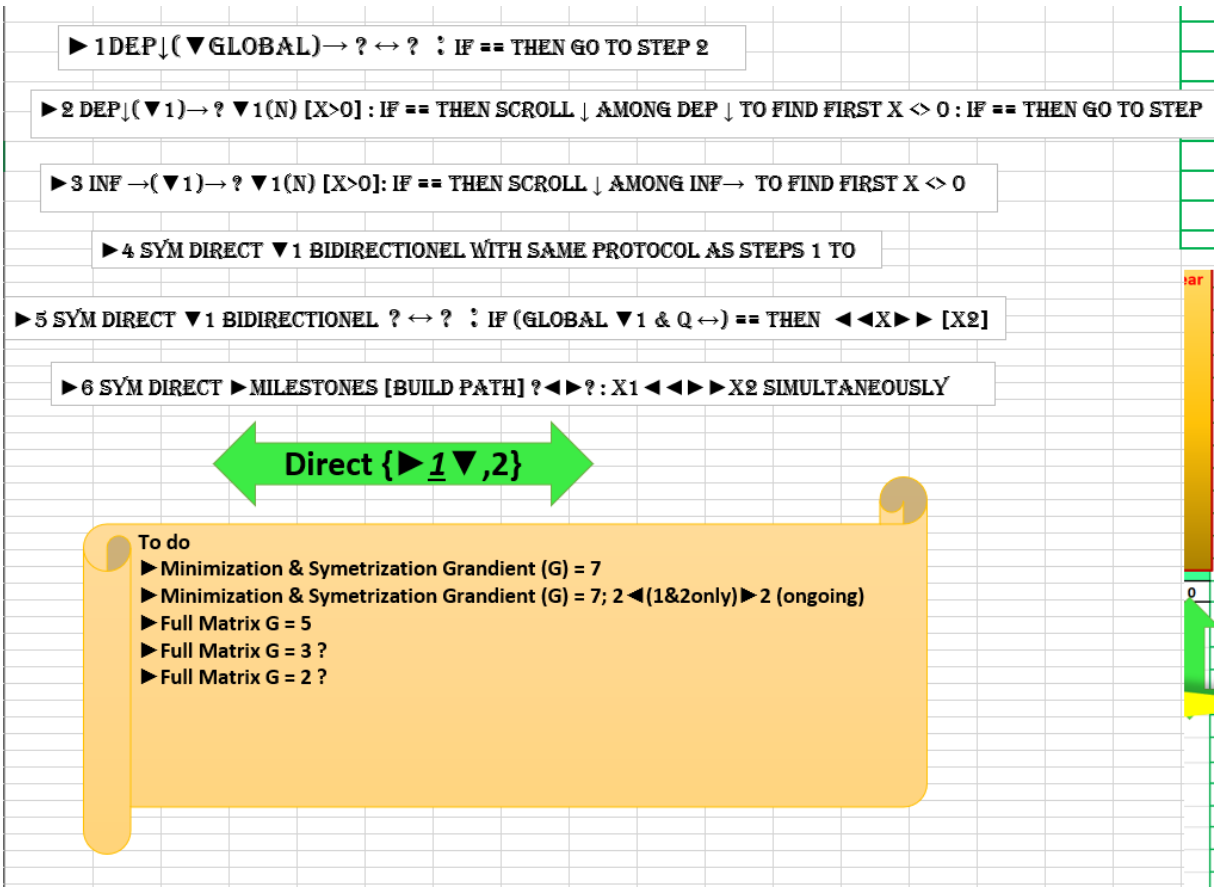
To do
►Minimization & Symetrization Grandient (G) = 7
►Minimization & Symetrization Grandient (G) = 7; 2◄(1&2only)►2 (ongoing)
►Full Matrix G = 5
►Full Matrix G = 3 ?
►Full Matrix G = 2 ?

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS STEPS 1 TO 3

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN ◄◄X►► [X2]

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2 SIMULTANEOUSLY

03/05/2023 11:59:38



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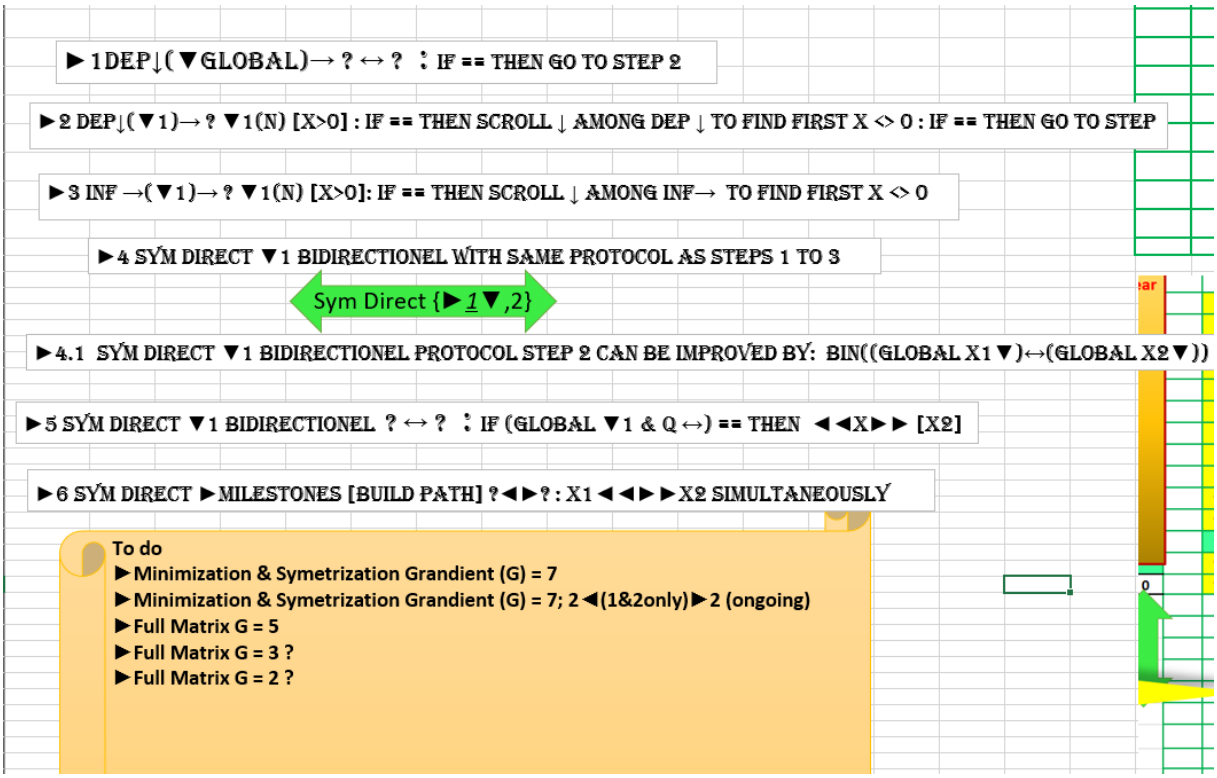
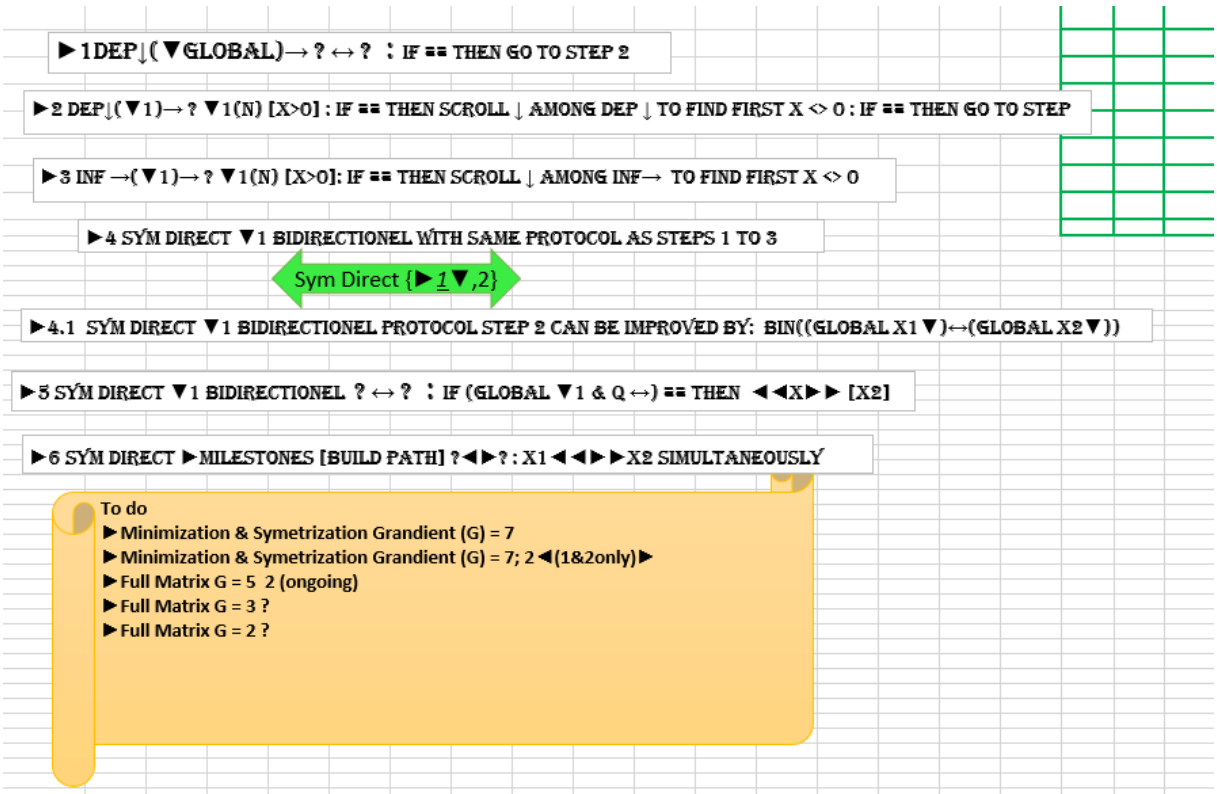


04/05/2023 18:29:55



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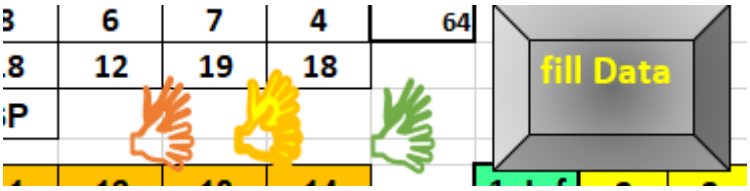
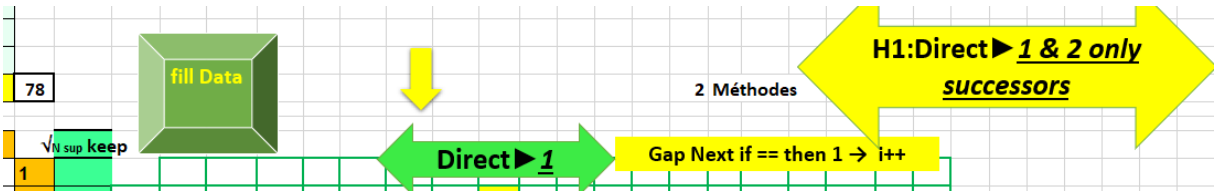
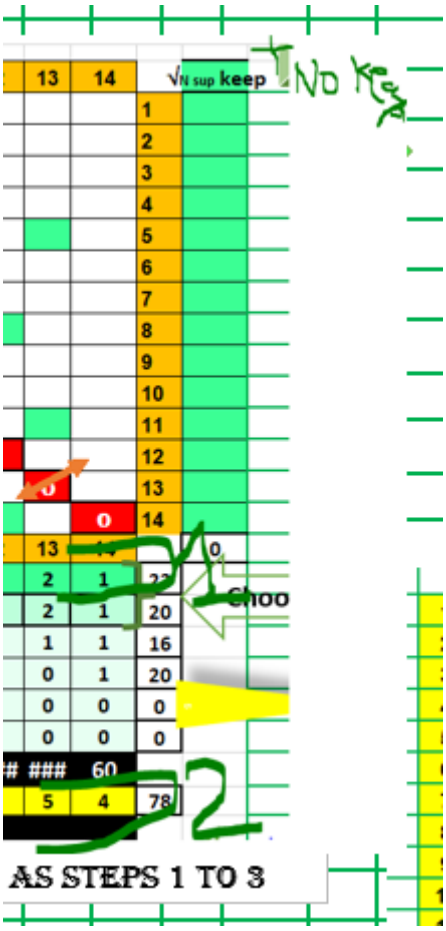


08/05/2023 20:23:28





+ roadmap

08/05/2023 23:07:37

				∑▼1	[1111]
			Direct 1		
				∑▼Global	[1112]
		√ sup keep			
				∑▼1	[1121]
			1 & 2 only successors		
				∑▼Global	[1122]
	√ n				
				∑▼1	[1231]
			Direct 1		
				∑▼Global	[1232]
		√ only Inf → Min			
				∑▼1	[1241]
			1 & 2 only successors		
				∑▼Global	[1242]

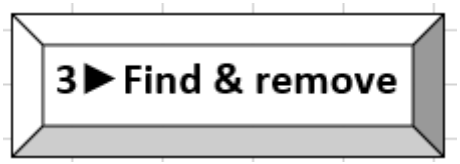
09/05/2023 20:22:07

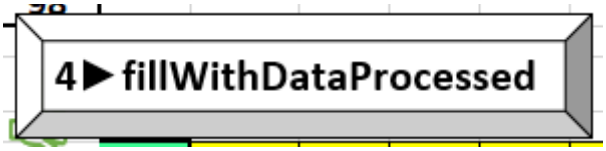
	Current heuristic			
			∑▼1	[1111]
		Direct 1		
			∑▼Global	[1112]
	Gradient sup keep all inf→ + dep↓			
			∑▼1	[1121]
		1 & 2 only successors		
			∑▼Global	[1122]
Gradient▶7				
			∑▼1	[1231]
		Direct 1		
			∑▼Global	[1232]
	Gradient only Inf → Min			
			∑▼1	[1241]
		1 & 2 only successors		
			∑▼Global	[1242]

10/05/2023 20:32:32

| | | |

12/05/2023 20:26:48





13/05/2023 16:45:48

			∑▼1		[111]
		Direct 1			
				Nodes D	[1121]
			∑▼Global		
				∑ nodes(I+D)	[1122]
1	Gradient 7				
		∑	I+D	+ Chirality	[121]
		∑ Ord D			[13]
		∑ Ord (D+I)			[14]

13/05/2023 17:42:05

roadMapfullMatrix					
				Direct	[11111]
			Forward→		
				+ Chirality	[11112]
		∑▼1			
			Bidirectional ↔	[1112]	
	Direct 1				
			Nodes D	[1121]	
		∑▼Global			
			∑ nodes(I+D)	[1122]	
1 Gradient 7					
	∑ I+D	+ Chirality	[121]		
	∑ Ord D	[13]			
	∑ Ord (D+I)	[14]			

13/05/2023 20:50:29

roadMapfullMatrix							
					Direct		[11111]
				Forward→			
					+ Chirality	if Stationary	[11112]
			∑▼1				
				Bidirectional ↔	[1112]		
		Direct 1					
				Nodes D	[1121]		
			∑▼Global				
				∑ nodes(I+D)	[1122]		
1	Gradient 7						
		∑ I+D	+ Chirality	[121]			
		∑ Ord D	[13]				
		∑ Ord (D+I)	[14]				

14/05/2023 18:43:31

Stationary Point	SP				
Local Minimum	LM				
Saddle Point	SD				
Chiral Points	CP				
Dirac Number	DN				

14/05/2023 19:58:36

►1DEP↓(▼GLOBAL)→ ? ↔ ? : IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL ↓ AMONG DEP ↓ TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL ↓ AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;
2►(X,Y=(1,>1)→CHIRAL (OK);3►(X,Y= >1,>1)→SUM NODES D

To do
►Minimization & Symetrization Grandient (G) = 7
►Minimization & Symetrization Grandient (G) = 7;
2◄(1&2only)►
►Full Matrix G = 7 (ongoing)
►Full Matrix G = 5
►Full Matrix G = 3 ?
►Full Matrix G = 2 ?

14/05/2023 20:14:55

0	0	0	0	0	0	
0	0	0	0	0	0	
0	0	0	0	0	0	
0	0	0	0	0	0	
0	0	0	0	0	0	2
0	0	0	0	0	0	4
0	0	0	0	0	0	
0	0	0	0	0	0	
9	10	11	12	13	14	

ALGO
Order
Gradient [▼1↔, ▼↕]
chiral
Stationary Point
bidirectional
prevent collision protocol [consequent]

►7 HOW TO CHOOSE IN FULL MATRIX IF :
1►(X,Y=1)→SUM▼1; 2►(X,Y=(1,>1)→CHIRAL (OK);3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP				
Local Minimum	LM	(2,4)	(5,13)	(8,12)	
Saddle Point	SD	(8,13)	(2,12)		
Chiral Points	CP	(3,5)			
Dirac Number	DN				

9	10	11	12	13	14		1_Inf

14/05/2023 22:29:38

►1DEP↓(▼GLOBAL)→ ? ↔ ? ⁚ IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL ↓ AMONG DEP ↓ TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL ↓ AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? ⁚ IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;1.1►IF LM → NEXT ONE →SUM NODES D 2►(X,Y=(1,>1)→CHIRAL (OK);3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP
Local Minimum	LM
Saddle Point	SD
Chiral Points	CP
Dirac Number	DN

To do
►Minimization & Symetrization Grandient (G) = 7
►Minimization & Symetrization Grandient (G) = 7;
2◄(1&2only)►
►Full Matrix G = 7 (ongoing)
►Full Matrix G = 5
►Full Matrix G = 3 ?
►Full Matrix G = 2 ?

14/05/2023 22:47:37

►1DEP↓(▼GLOBAL)→ ? ↔ ? : IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL | AMONG DEP | TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL | AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;1.1►IF EQUAL →SUM NODES D; 1.2►IF LM → NEXT ONE →SUM NODES D || 2►(X,Y=(1,>1)→CHIRAL (OK) ||3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP
Local Minimum	LM
Saddle Point	SD
Chiral Points	CP
Dirac Number	DN

To do
►Minimization & Symetrization Grandient (G) = 7
►Minimization & Symetrization Grandient (G) = 7;
2◄(1&2only)►
►Full Matrix G = 7 (ongoing)
►Full Matrix G = 5
►Full Matrix G = 3 ?
►Full Matrix G = 2 ?

15/05/2023 12:10:36

►1DEP↓(▼GLOBAL)→ ? ↔ ? : IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL ↓ AMONG DEP ↓ TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL ↓ AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;1.1►IF EQUAL →SUM NODES D; 1.2►IF LM → NEXT ONE →SUM NODES D || 2►(X,Y=(1,>1)→CHIRAL (OK) ||3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP
Local Minimum	LM
Saddle Point	SD
Chiral Points	CP
Dirac Number	DN

►8 INF→(▼1) > DEP(▼1)↓

14	2	3		6	
	1	2	3	4	5
1_Dep ↓	0	1	1	2	2
1_Inf→ >	1	1	1	1	1

To do
- ►Minimization & Symetrization Grandient (G) = 7
- ►Minimization & Symetrization Grandient (G) = 7;
2◄(1&2only)►
- ►Full Matrix G = 7 (ongoing)
- ►Full Matrix G = 5
- ►Full Matrix G = 3 ?
- ►Full Matrix G = 2 ?

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roadMapfullMatrix

To do
- ►Minimization & Symetrization Grandient (G) = 7
- ►Minimization & Symetrization Grandient (G) = 7; 2◄(1&2only)►
- ►Full Matrix G = 7 (ongoing)
- ►Full Matrix G = 5
- ►Full Matrix G = 3 ?
- ►Full Matrix G = 2 ?

Forward→ — Direct [11111]

+ Chirality if Stationary [11112]

∑▼1

Bidirectional ↔ [1112]

Direct 1

↔ Nodes D [1121]

∑▼Global

↔ ∑ nodes(I+D) [1122]

1 Gradient 7

↔ ∑ I+D + Chirality [121]

↔ ∑ Ord D [13]

↔ ∑ Ord (D+I) [14]

[11111] ∑▼1

1	11	6	9	10	3	5	13	8	12	
2	4	7	8	12						
3	5	13	11							
4	2	12	8							
5	3	10	9	6	11	1	14	12	?	
6	9	?								
7	4	2	?							
8	12									
9	6	11	1	14	12	8				
10	3	5	13	11						
11	1	14	12	8						
12	8									
13	5	3	10	9	6	11	1	14	12	8
14	12	8								

[11112] ∑▼1 + Chirality If Stationary + Nodes D

1	11	6	9	10	3	5	13	8	14					
2	4	7	12											
3	10	9	6	11	1	14	12	2	4	7	8	13	5	3
4	7	2												
5	3	10	9	6	11	1	14	12	2	4	7	8	13	5
6	9	10	3	5	13	11								
7	4	2	14											
8	12													
9	6	11	1	14	12	2	4	7	8	13	5	3	10	9
10	3	5	13	11										
11	1	14	12	2	4	7	8	13	5	3	10	6		
12	14	1	11	6	9	10	3	5	13	8	7	4	2	12
13	5	3	10	9	6	11	1	14	12	2	4	7	8	13
14	12	2	4	7	8	13	5	3	10	9	6	11	1	14

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►1DEP↓(▼GLOBAL)→ ? ↔ ? : IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL ↓ AMONG DEP ↓ TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL ↓ AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;1.1►IF EQUAL →SUM NODES D; 1.2►IF LM → NEXT ONE →SUM NODES D || 2►(X,Y=(1,>1)→CHIRAL (OK) ||3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP
Local Minimum	LM
Saddle Point	SD
Chiral Points	CP
Dirac Number	DN

►8 INF→(▼1) > DEP(▼1)↓

14	2	3		6	
	1	2	3	4	5
1_Dep ↓	0	1	1	2	2
1_Inf→ >	1	1	1	1	1

►9 CASE : ↔ SUB MATRIX SCROLL TO KNOW WHICH DIRECTION TO TAKE: ?◄►?

14	2	3		6			7
	1	2	3	4	5	6	7
1_Dep ↓	0	1	1	2	2	1	0
1_Inf→ >	1	1	1	1	1	1	1
2	4	1	0	0	0	2	1
3	0	1	1	0	1	1	1
4	1	1	2	0	2	1	0
5	1	0	1	0	1	2	0
6	4	0	0	1	2	0	2
7	3	0	3	1	0	0	1

To do
- ►Minimization & Symetrization Grandient (G) = 7
- ►Minimization & Symetrization Grandient (G) = 7; 2◄(1&2only)►
- ►Full Matrix G = 7 (ongoing)
- ►Full Matrix G = 5
- ►Full Matrix G = 3 ?
- ►Full Matrix G = 2 ?

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►1 DEP↓(▼GLOBAL)→ ? ↔ ? : IF == THEN GO TO

►2 DEP↓(▼1)→ ? ▼1(N) [X>0] : IF == THEN SCROLL ↓ AMONG DEP ↓ TO FIND FIRST X <> 0 : IF ==

►3 INF →(▼1)→ ? ▼1(N) [X>0]: IF == THEN SCROLL ↓ AMONG INF→ TO

►4 SYM DIRECT ▼1 BIDIRECTIONEL WITH SAME PROTOCOL AS

Sym Direct

►4.1 SYM DIRECT ▼1 BIDIRECTIONEL PROTOCOL STEP 2 CAN BE IMPROVED BY: BIN((GLOBAL

►5 SYM DIRECT ▼1 BIDIRECTIONEL ? ↔ ? : IF (GLOBAL ▼1 & Q ↔) == THEN

►6 SYM DIRECT ►MILESTONES [BUILD PATH] ?◄►? : X1◄◄►►X2

►7 HOW TO CHOOSE IN FULL MATRIX IF : 1►(X,Y=1)→SUM▼1;1.1►IF EQUAL →SUM NODES D; 1.2►IF LM → NEXT ONE →SUM NODES D || 2►(X,Y=(1,>1)→CHIRAL (OK) ||3►(X,Y= >1,>1)→SUM NODES D

Stationary Point	SP
Local Minimum	LM
Saddle Point	SD
Chiral Points	CP
Dirac Number	DN

►8 INF→(▼1) > DEP(▼1)↓

14	2	3		6	
	1	2	3	4	5
1_Dep ↓	0	1	1	2	2
1_Inf→ >	1	1	1	1	1

►9 CASE : ↔ SUB MATRIX SCROLL TO KNOW : 1. WICH DIRECTION TO TAKE: ?◄►? ; 2. WICH NEXT VERTICE TO TAKE

14	2	3		6			7
	1	2	3	4	5	6	7
1_Dep ↓	0	1	1	2	2	1	0
1_Inf→ >	1	1	1	1	1	1	1
2	4	1	0	0	0	2	1
3	0	1	1	0	1	1	1
4	1	1	2	0	2	1	0
5	1	0	1	0	1	2	0
6	4	0	0	1	2	0	2
7	3	0	3	1	0	0	1

To do
- ►Minimization & Symetrization Grandient (G) = 7
- ►Minimization & Symetrization Grandient (G) = 7; 2◄(1&2only)►
- ►Full Matrix G = 7 (ongoing)
- ►Full Matrix G = 5
- ►Full Matrix G = 3 ?
- ►Full Matrix G = 2 ?

End